



**DEPARTMENT OF MATHEMATICS AND COMPUTING**  
**NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA**  
(An Institution of National Importance under the Ministry of Education, Govt. of India)  
**END SEMESTER EXAMINATIONS, AUGUST 2026**

**M&C IV Semester**

**MAPC-206: Theory of Computation**

**Date:** 11-08-2026

**Time:** 02:00 PM to 05:00 PM

**Max. Marks:** 50

**N.B.:** Answer ALL questions.

Answers to all parts of each question should be in one place.

|           |                                                                                                                                                                                                                                                                                                         |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1</b> | Define Automata and Grammar. Write powers of sigma along with Kleene closure and positive closure. [6M]                                                                                                                                                                                                 |
| <b>Q2</b> | Define deterministic finite Automata (DFA). Construct a DFA for a string start with $a$ over a given alphabet $\Sigma = \{a, b\}$ . [6M]                                                                                                                                                                |
| <b>Q3</b> | Construct a DFA which accepts a language of all strings starting with symbol $a$ and ending with symbol $b$ over the alphabet $\Sigma = \{a, b\}$ . [6M]                                                                                                                                                |
| <b>Q4</b> | Write difference between Non deterministic finite Automata (NFA) and DFA. Construct a NFA of all binary strings in which second last symbol is 1 over the alphabet $\Sigma = \{0, 1\}$ . [6M]<br>-OR-<br>What is halting problem? Prove that halting problem is an undecidable problem in general. [6M] |
| <b>Q5</b> | Convert the NFA defined in the above question into DFA. [6M]<br>-OR-<br>Define Moore machine and Mealy machine with one example of your choice in each case. [6M]                                                                                                                                       |
| <b>Q6</b> | Define quotient operator, INIT operation, homomorphism and inverse homomorphism with at least one example in each case. [6M]<br>-OR-<br>Define push down Automata (PDA). How to know whether a string is finally accepted by a PDA? Explain it in detail with at least one example. [6M]                |
| <b>Q7</b> | Define Turing machine with one example of your choice. [6M]<br>-OR-<br>Define the universal Turing machine. Write difference between recursive enumerable language (REL) and recursive language (RL). [6M]                                                                                              |
| <b>Q8</b> | I. Explain Chomsky normal form (CNF). Convert the following context free grammar (CFG) into CNF: $P: S \rightarrow ASA/aB, A \rightarrow B/S, B \rightarrow b/\epsilon$ . [5M]<br>II. Explain post correspondence problem (PCP) as an undecidable problem with one example. [3M]                        |

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